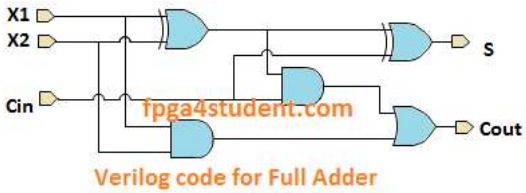
Verilog code for Full Adder

In this Verilog project, Verilog code for Full Adder is presented. Both behavioral and structural Verilog code for Full Adder is implemented.



Verilog code for the full adder using behavioral code:

```
 // fpga4student.com
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// Verilog code for full adder
// Behavioral code for full adder
module Full_Adder_Behavioral_Verilog(
  input X1, X2, Cin,
  output S, Cout
  );
    reg[1:0] temp;
   always @(*)
   begin
   temp = {1'b0,X1} + {1'b0,X2}+{1'b0,Cin};
   end
   assign S = temp[0];
   assign Cout = temp[1];
endmodule
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// Verilog code for full adder
// Testbench code of the behavioral code for full adder
`timescale 10ns/ 10ps;
module Testbench_Behavioral_adder();
 reg A,B,Cin;
 wire S,Cout;
 //Verilog code for the structural full adder
 Full_Adder_Behavioral_Verilog Behavioral_adder(
    .X1(A),
    .X2(B),
    .Cin(Cin),
    .S(S),
    .Cout(Cout)
   );
 initial begin
   A = 0;
   B = 0;
   Cin = 0;
   #5;
   A = 0;
   B = 0;
   Cin = 1;
   #5;
   A = 0;
   B = 1;
   Cin = 0;
   #5;
   A = 0;
   B = 1;
```
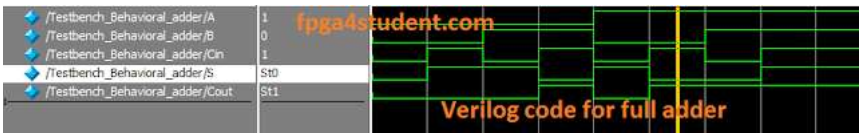

```
    Cin = 1;
    #5;
    A = 1;
    B = 0;
    Cin = 0;
    #5;
    A = 1;
    B = 0;
    Cin = 1;
    #5;
    A = 1;
    B = 1;
    Cin = 0;
    #5;
    A = 1;
    B = 1;
    Cin = 1;
    #5;
   end

 endmodule
```

Simulation wave of the behavioral Verilog code for the full adder:



Verilog code for the full adder using structural code:

```
// fpga4student.com
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// Verilog code for full adder
// Structural code for full adder
module Full_Adder_Structural_Verilog(
  input X1, X2, Cin,
  output S, Cout
  );
    wire a1, a2, a3;
    xor u1(a1,X1,X2);
 and u2(a2,X1,X2);
 and u3(a3,a1,Cin);
 or u4(Cout,a2,a3);
    xor u5(S,a1,Cin);
endmodule
// fpga4student.com
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// Verilog code for full adder
// Testbench code of the structural code for full adder
`timescale 10ns/ 10ps;
module Testbench_structural_adder();
 reg A,B,Cin;
 wire S,Cout;
 //Verilog code for the structural full adder
 Full_Adder_Structural_Verilog structural_adder(
    .X1(A),
    .X2(B),
    .Cin(Cin),
    .S(S),
    .Cout(Cout)
   );
 initial begin
   A = 0;
   B = 0;
   Cin = 0;
   #10;
   A = 0;
   B = 0;
   Cin = 1;
```
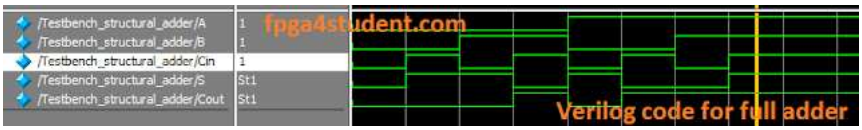

```
    #10;
    A = 0;
    B = 1;
    Cin = 0;
    #10;
    A = 0;
    B = 1;
    Cin = 1;
    #10;
    A = 1;
    B = 0;
    Cin = 0;
    #10;
    A = 1;
    B = 0;
    Cin = 1;
    #10;
    A = 1;
    B = 1;
    Cin = 0;
    #10;
    A = 1;
    B = 1;
    Cin = 1;
    #10;
  end

endmodule
```

Simulation wave of the structural Verilog code for the full adder:



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26. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-2)
27. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-3)

28. Verilog code for Decoder
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